



TOPIC 1

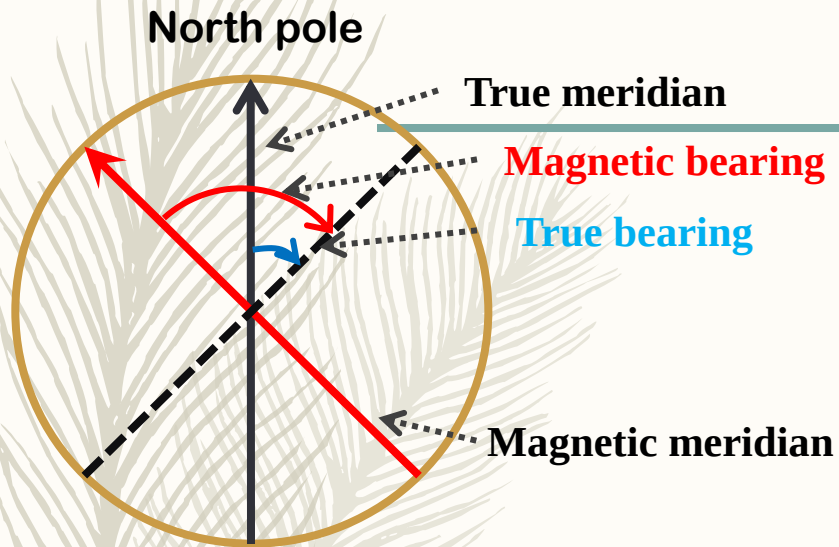
BEARING
AND
ANGLE

OBJECTIVE

End of these topic, student should be able to:

1. Differentiate terms of bearing and angle.
2. Differentiate terms of quadrant bearing and whole circle bearing.
3. Demonstrate whole circle bearing conversion to quadrant bearing.
4. Calculate the internal angle from two bearings.
5. Calculate the bearing from internal angle and a known bearing.

IMPORTANT DEFINITIONS

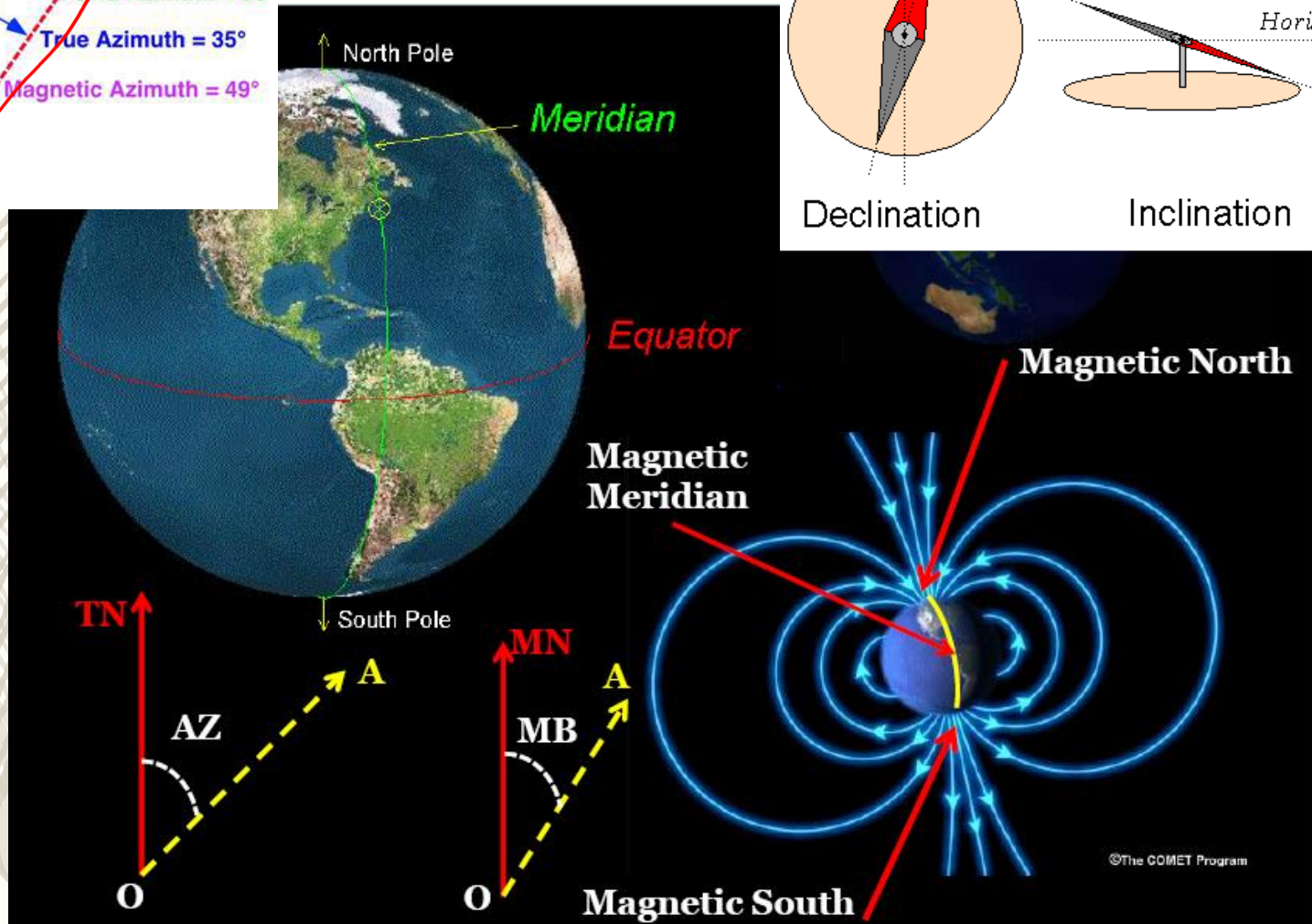
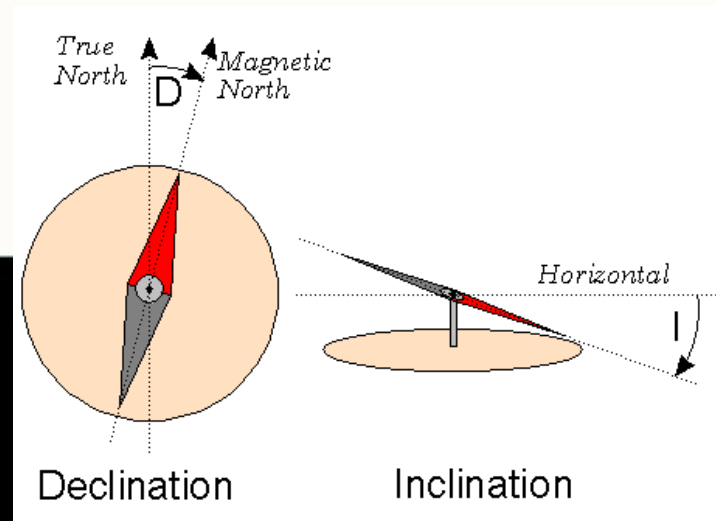
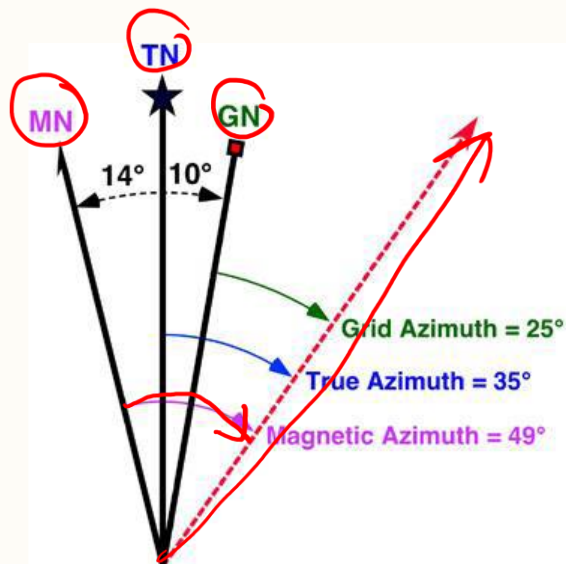


True meridian:

Line or plane passing through geographical north pole and geographical south pole

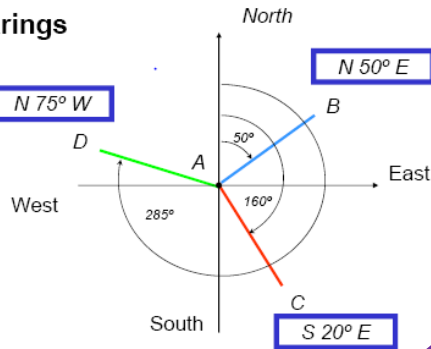
Magnetic meridian:

- The direction indicated by a freely suspended and properly balanced magnetic needle is known as Magnetic Meridian.
- The angle between the magnetic meridian and a line is known as magnetic bearing or simple bearing of the line.



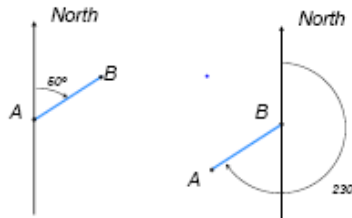
BEARING vs ANGLE

Bearings

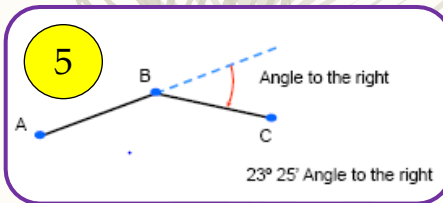
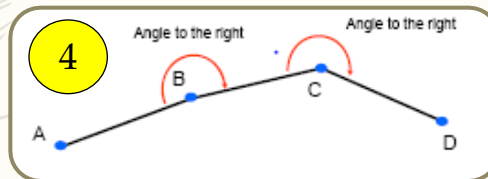
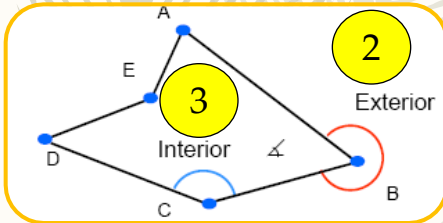


BEARING

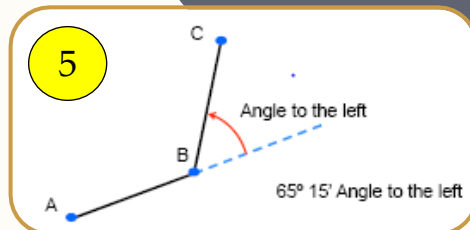
For example: the *forward azimuth* of line AB is 50° - the *back azimuth* or azimuth of BA is 230°



1. A bearing is a direction
2. Another method of describing the direction of a line
3. Measured clockwise from north
4. **North** is 0° , **East** is 90° , **South** is 180° , and **West** is 270°
5. Using symbol = Degree ($^\circ$) Minute ($'$) Second ($''$)
6. **TWO** types :
 - i. Quadrant bearing
 - ii. Whole Circle Bearing



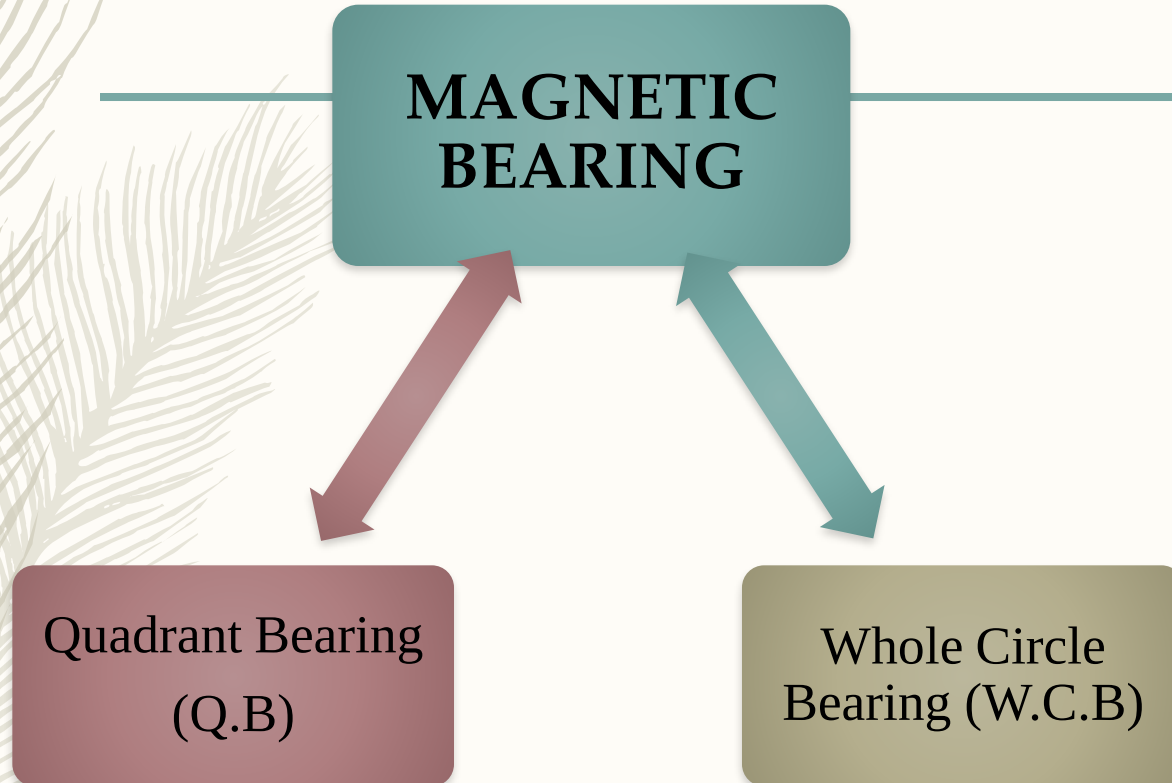
ANGLE



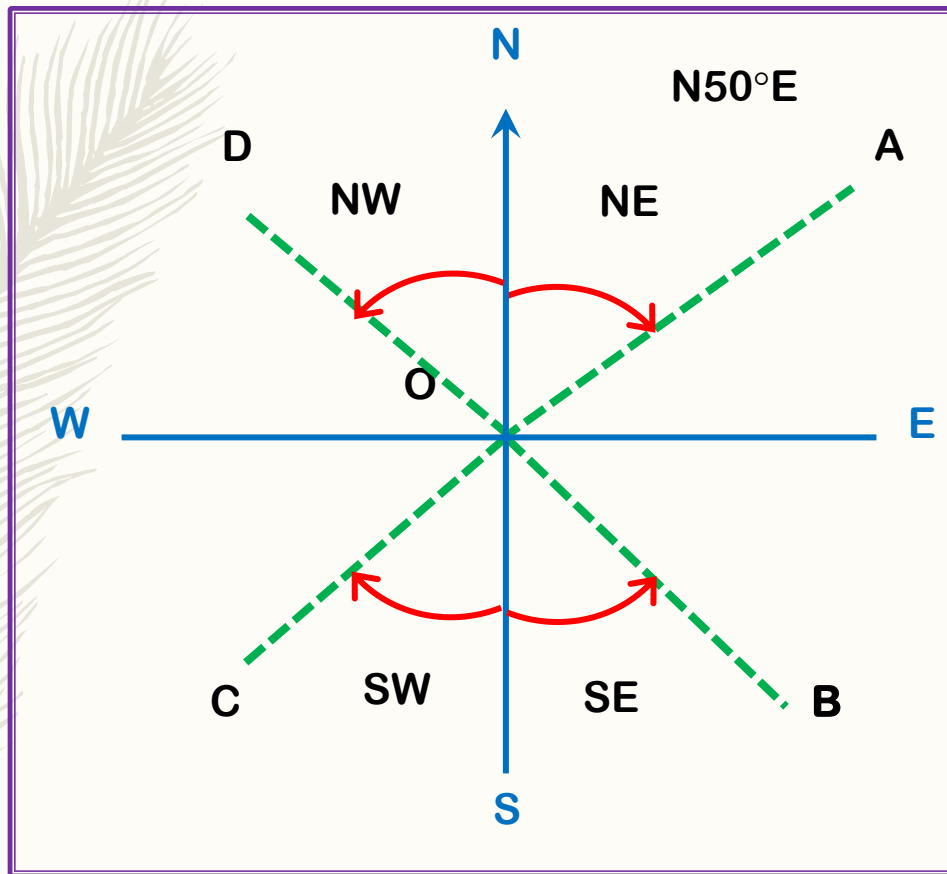
1. Measure of rotation from one ray to another ray with the same endpoint.
2. An **exterior angle** is one that is not enclosed by the sides of a closed traverse
3. An **interior angle** is one enclosed by sides of a closed traverse
4. An **angle to the right** is the clockwise angle between the preceding line and the next line of the a traverse
5. A **deflection angle** the angle between the preceding line and the present one

BEARING

There are TWO (2) types of Magnetic bearing:



QUADRANT BEARING (Q.B)

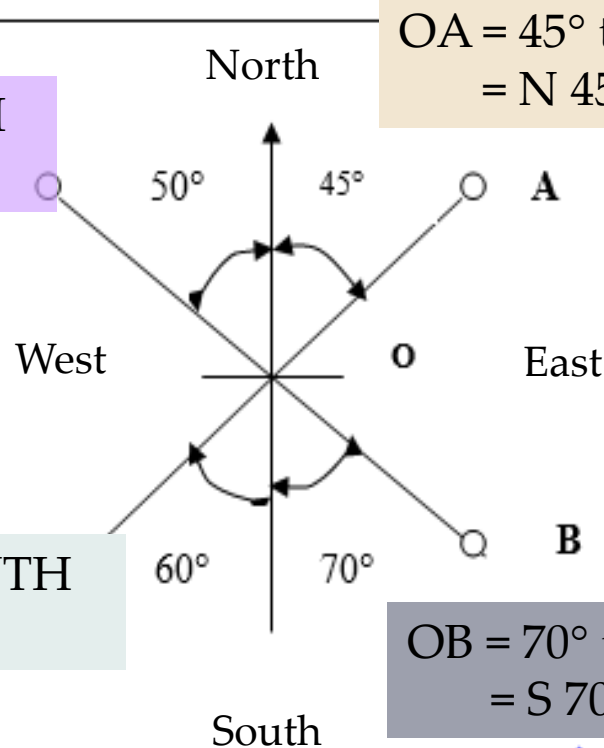


- The bearing of survey lines are measured from **O** to **north line** or **south line** which ever is the nearest to the given survey line.
- Either in clockwise direction or in anti clockwise direction. This system consists of **4-quadrants NE, SE, NW, SW**.

OD = 50° to WEST from NORTH
= N 50° W

D

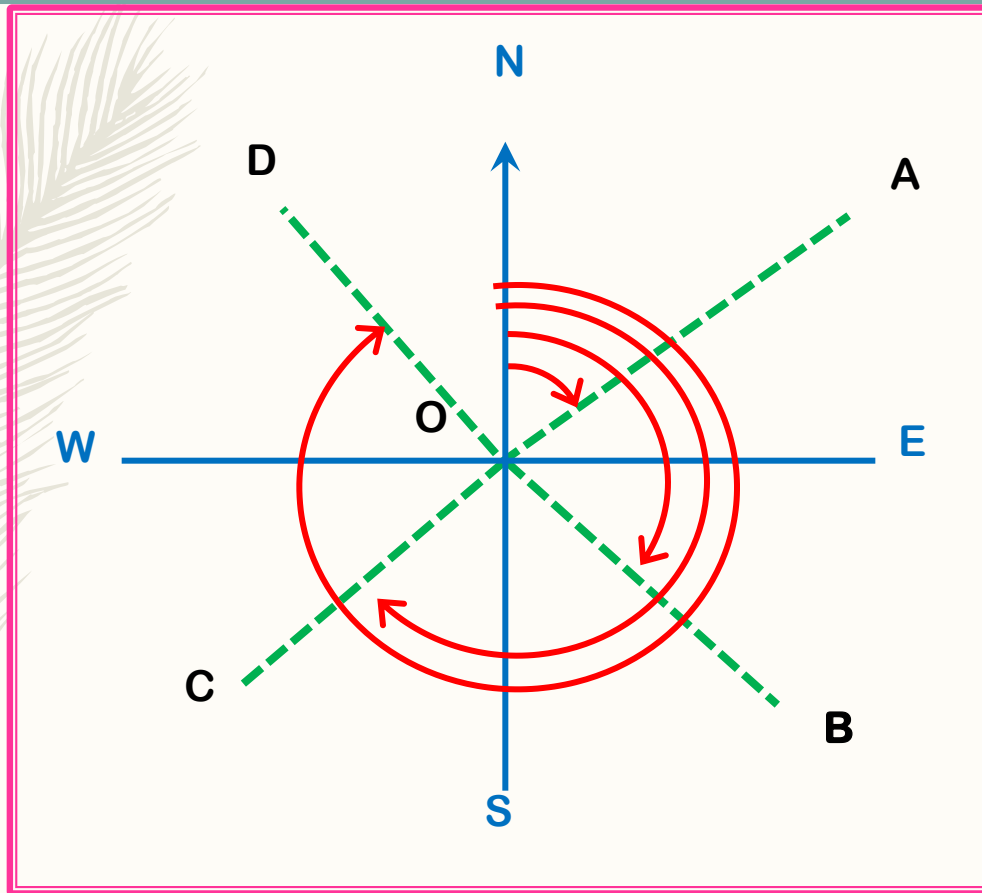
OC = 60° to WEST from SOUTH
= S 60° W



OA = 45° to EAST from NORTH
= N 45° E

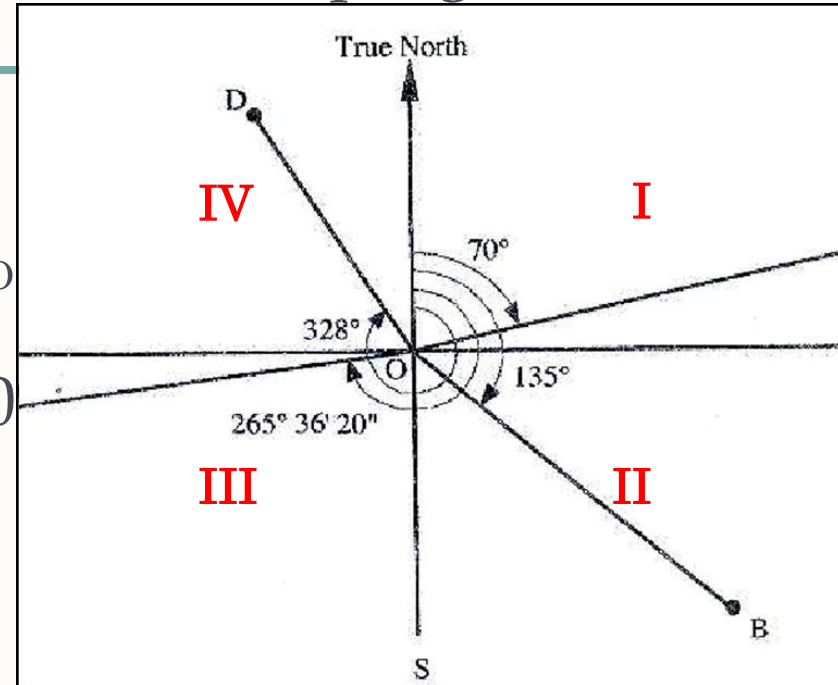
OB = 70° to EAST from SOUTH
= S 70° E

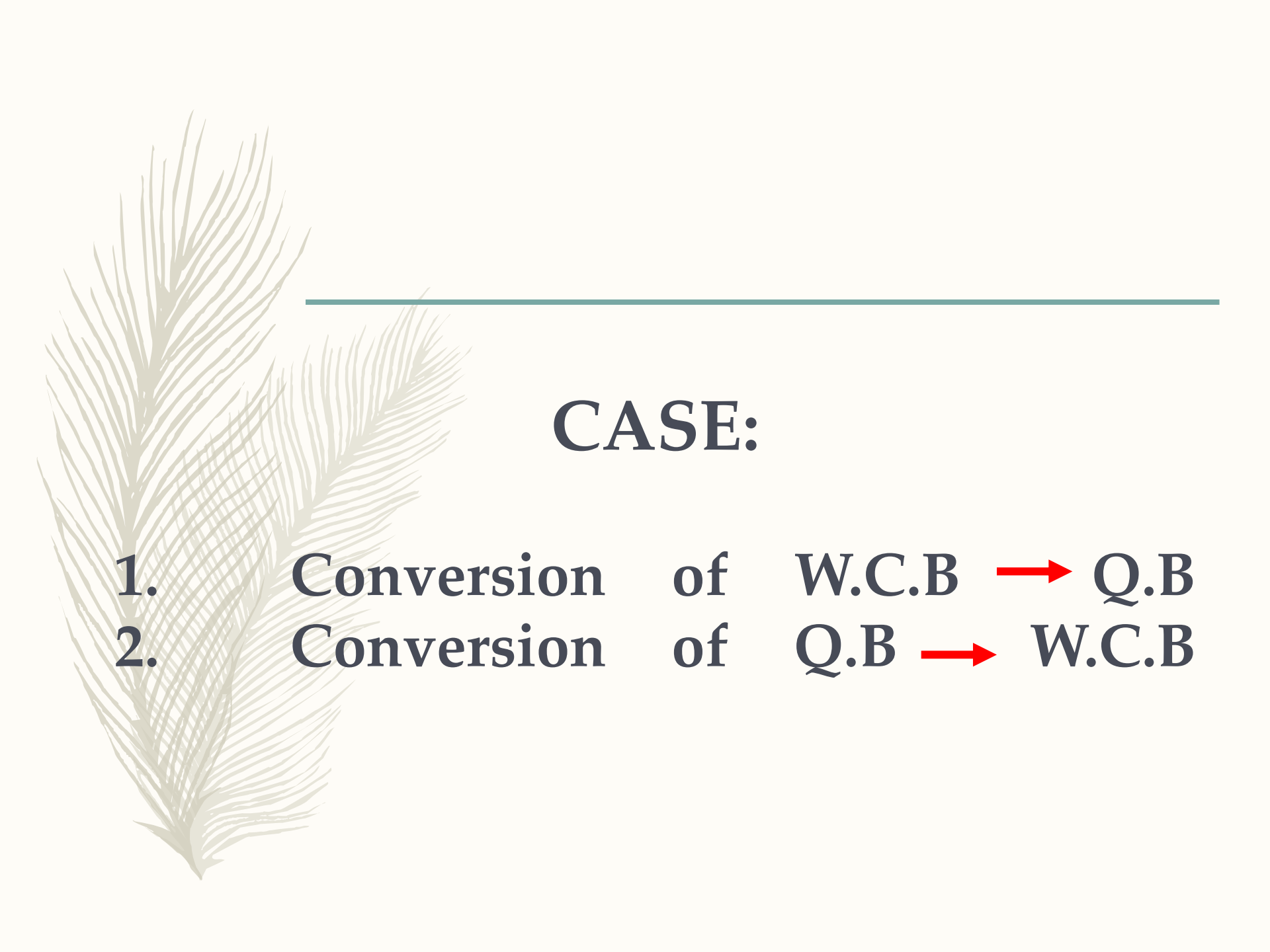
WHOLE CIRCLE BEARING (W.C.B)



- The magnetic bearing of a line measured clockwise from the North Pole towards the line
- The quadrant start from north and progress in a clockwise direction as:

- 1) 1st quadrant is 0° to 90°
- 2) 2nd quadrant is 90° to 180°
- 3) 3rd quadrant is 180° to 270°
- 4) 4th quadrant is up to 360°

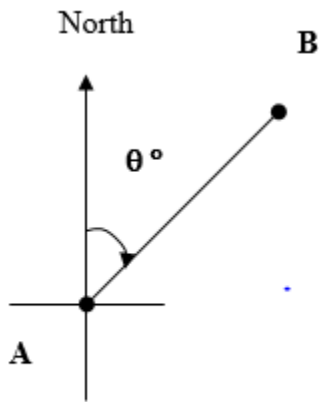




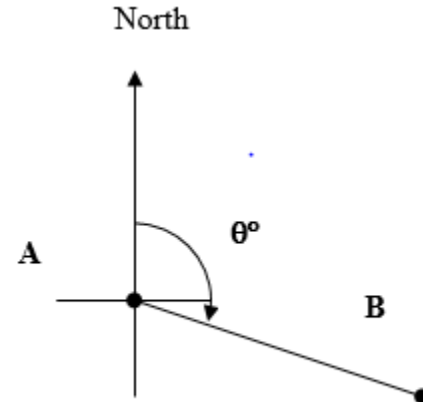
CASE:

1. Conversion of W.C.B $\rightarrow$ Q.B
2. Conversion of Q.B $\rightarrow$ W.C.B

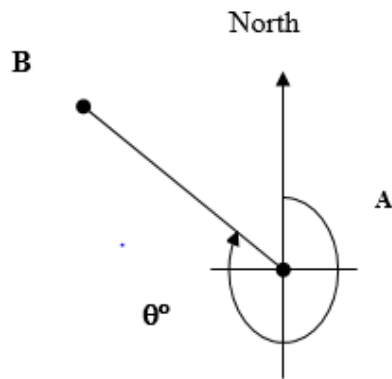
CASE 1: Conversion of W.C.B to Q.B



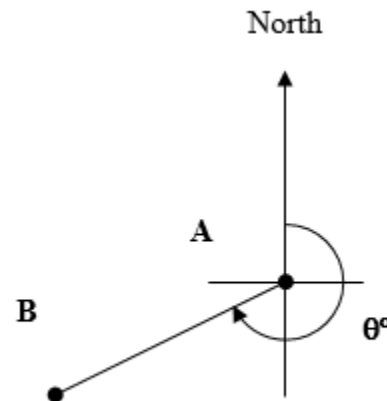
Quadrant Bearing A-B is N (θ°) E



Quadrant Bearing A-B is S ($180^\circ - \theta^\circ$) E



Quadrant Bearing A-B is N ($360^\circ - \theta^\circ$) W

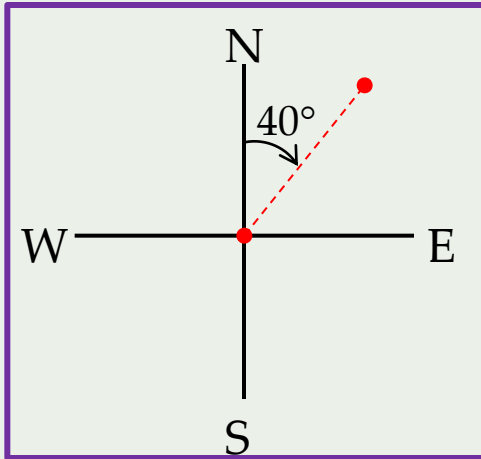


Quadrant Bearing A-B is S ($\theta^\circ - 180^\circ$) W

EXAMPLE:

Convert the following WCB into QB:

1) $40^{\circ} 00' 00''$

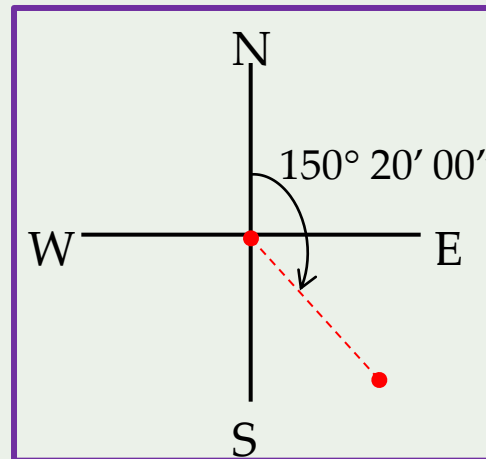


Answer:

WCB = $40^{\circ} 00' 00''$

QB = N $40^{\circ} 00' 00''$ E

2) $150^{\circ} 20' 00''$



Answer:

WCB = $150^{\circ} 20' 00''$

QB = $180^{\circ} - 150^{\circ} 20' 00''$

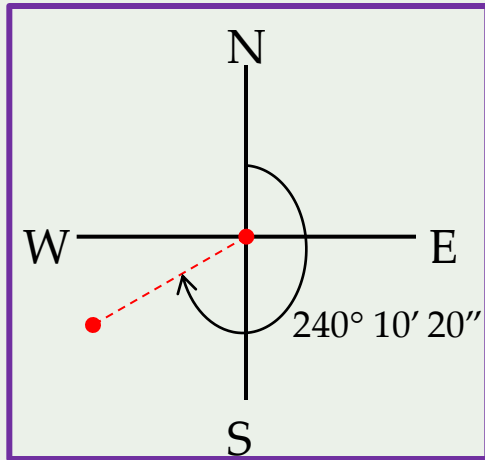
= $29^{\circ} 40' 00''$

= S $29^{\circ} 40' 00''$ E

EXAMPLE:

Convert the following WCB into QB:

3) $240^{\circ} 10' 20''$

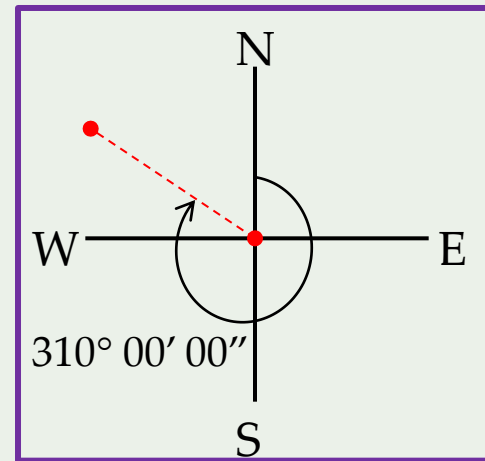


Answer:

$$\text{WCB} = 240^{\circ} 10' 20''$$

$$\begin{aligned}\text{QB} &= \theta - 180^{\circ} \\ &= 240^{\circ} 10' 20'' - 180^{\circ} \\ &= 60^{\circ} 10' 20'' \\ &= \underline{\underline{\text{S } 60^{\circ} 10' 20'' \text{ W}}}\end{aligned}$$

4) $310^{\circ} 00' 00''$



Answer:

$$\text{WCB} = 310^{\circ} 00' 00''$$

$$\begin{aligned}\text{QB} &= 360^{\circ} - \theta \\ &= 360^{\circ} - 310^{\circ} 00' 00'' \\ &= 50^{\circ} 00' 00'' \\ &= \underline{\underline{\text{N } 50^{\circ} 00' 00'' \text{ W}}}\end{aligned}$$

EXERCISE

1) Convert the following Whole Circle Bearing (WCB) to Quadrant Bearing (QB):

i. $342^{\circ} 54' 25''$

ii. $134^{\circ} 22' 50''$

iii. $274^{\circ} 15' 00''$

iv. $15^{\circ} 00' 20''$

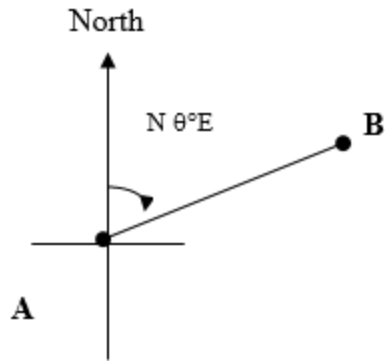
EXERCISE

- 2) Sketch position of the traverse and convert the following Whole Circle Bearing (WCB) to Quadrant Bearing (QB):

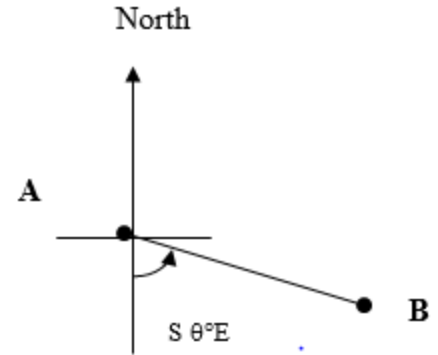


LINE	BEARING
1-2	10° 20' 30"
2-3	100° 02' 40"
3-4	210° 30' 50"
4-1	300° 00' 00"

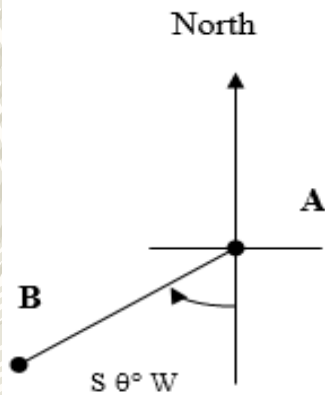
CASE 2: Conversion of Q.B to W.C.B



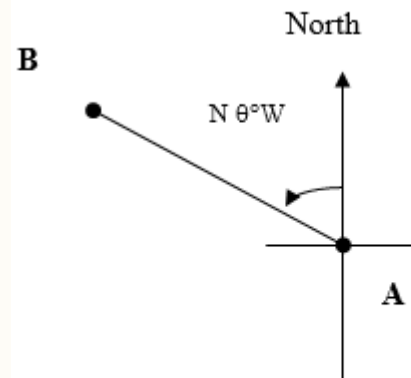
WC Bearing A-B is θ°



WC Bearing A-B is $180^\circ - \theta^\circ$



WC Bearing A-B is $180^\circ + \theta^\circ$

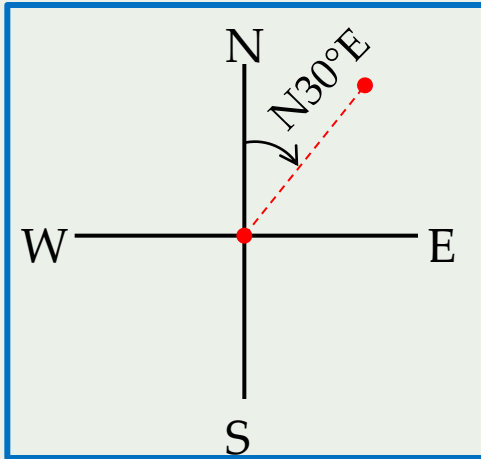


WC Bearing A-B is $360^\circ - \theta^\circ$

EXAMPLE:

Convert the following QB into WCB:

1) N 30° 00' 00'' E

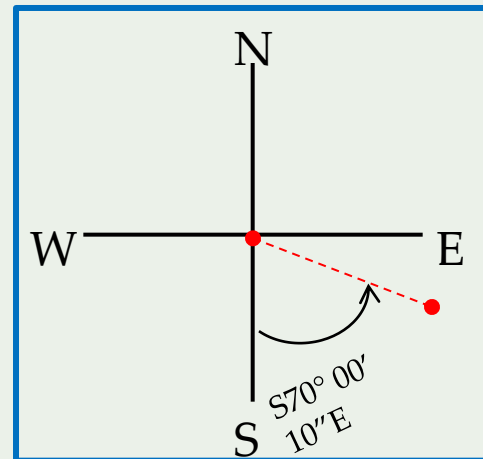


Answer:

QB = N 30° 00' 00'' E

WCB = 30° 00' 00''

2) S 70° 00' 10'' E



Answer:

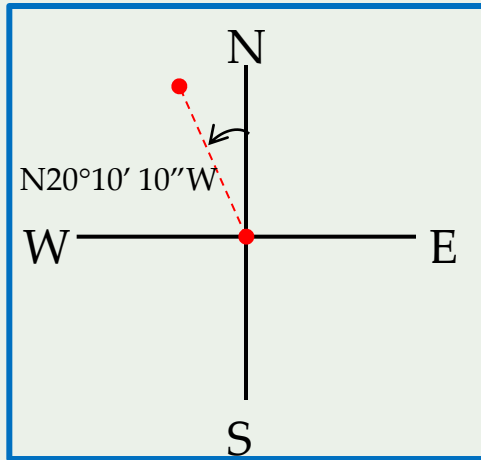
QB = S 70° 00' 10'' E

WCB = 180° - 70° 00' 10''
= 109° 59' 50''

EXAMPLE:

Convert the following QB into WCB:

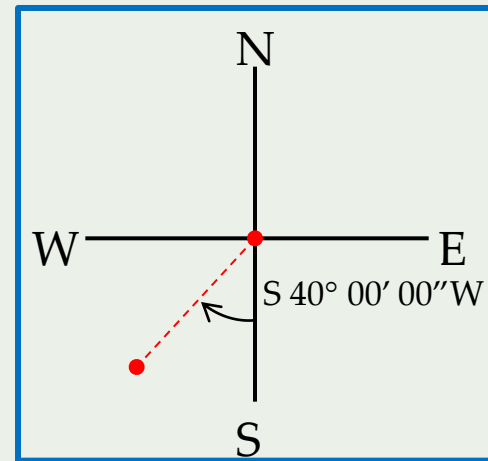
3) N 20° 10' 10'' W



Answer:

$$\begin{aligned}\text{QB} &= \text{N } 20^{\circ} 10' 10'' \text{ W} \\ \text{WCB} &= 360^{\circ} 00' 00'' - 20^{\circ} 10' 10'' \\ &= 339^{\circ} 49' 50'' \\ &= \underline{\underline{339^{\circ} 49' 50''}}\end{aligned}$$

4) S 40° 20' 00'' W



Answer:

$$\begin{aligned}\text{QB} &= \text{S } 40^{\circ} 20' 00'' \text{ W} \\ \text{WCB} &= 180^{\circ} + 40^{\circ} 20' 00'' \\ &= 220^{\circ} 20' 00'' \\ &= \underline{\underline{220^{\circ} 20' 00''}}\end{aligned}$$

EXERCISE

1) Convert the following Quadrant Bearing (QB) to Whole Circle Bearing (WCB) :

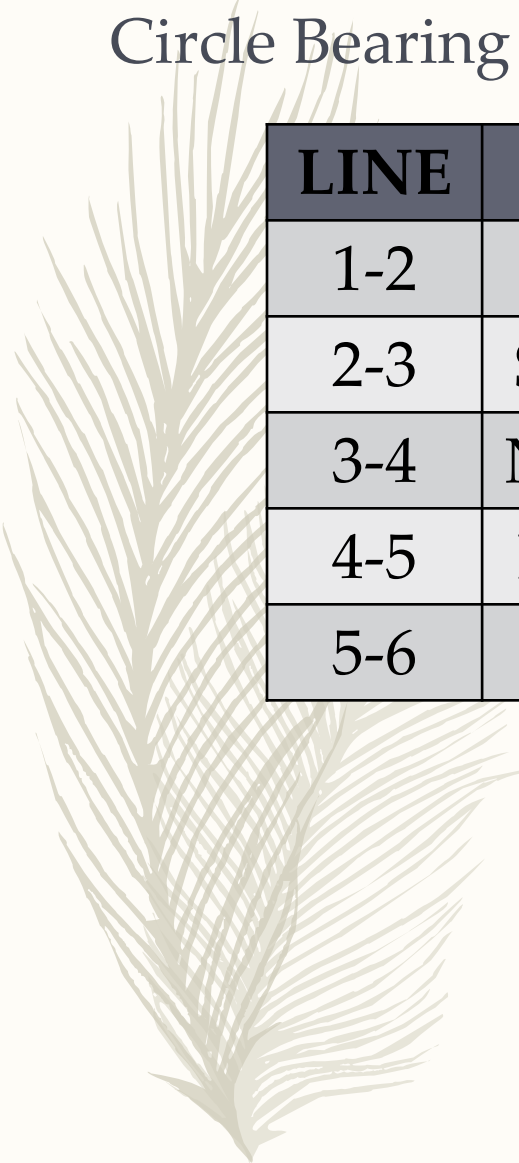
i. $S 20^{\circ} 20' 00'' E$

ii. $N 89^{\circ} 22' 50'' E$

iii. $N 32^{\circ} 52' 00'' W$

iv. $S 74^{\circ} 52' 00'' W$

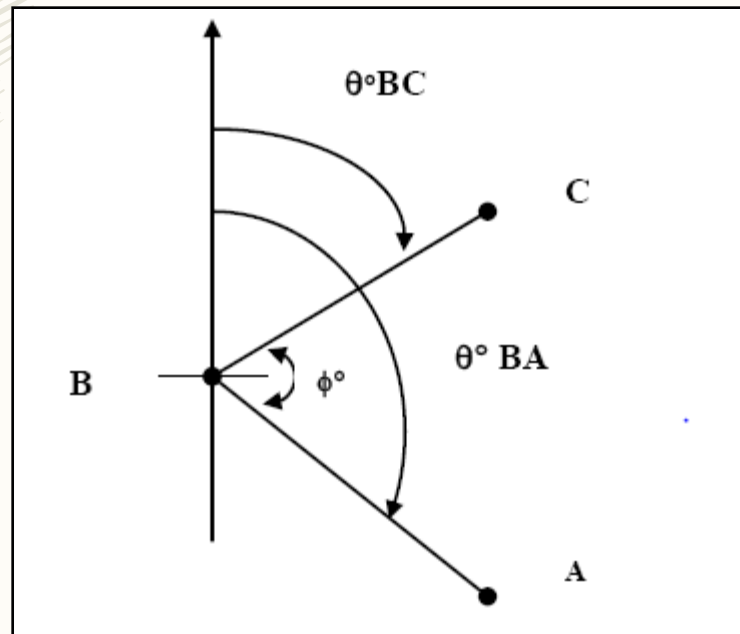
2) Convert the following Quadrant Bearing (QB) to Whole Circle Bearing (WCB)



LINE	QB	WCB
1-2	S 89° 59' 35" E	
2-3	S 42° 15' 30" W	
3-4	N 39° 29' 30" W	
4-5	N 88° 15' 45" E	
5-6	S 79° 59' 00" E	

INTERNAL ANGLE

Calculate The Inner Angle Of Two Value Bearings



- Each line traverse, the difference between forward bearings and back bearings is $\pm 180^\circ$.
- If back bearing is LESS than 180° , so to get forward bearing must be plus (+) with 180° .
- If back bearing is MORE than 180° , so to get forward bearing must be minus (-) with 180° .

EXAMPLE:

If Bearing AB = $112^\circ 15' 30''$

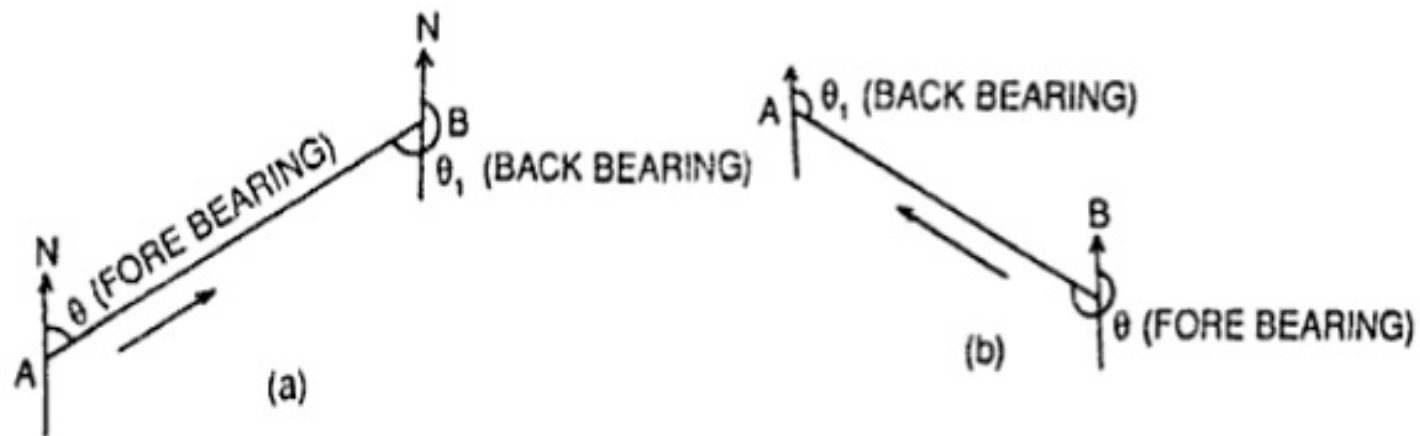
So,

$$\begin{aligned}\text{Bearing BA} &= 112^\circ 15' 30'' + 180^\circ 00' 00'' \\ &= 292^\circ 15' 30''\end{aligned}$$

If Bearing AB = $240^\circ 15' 30''$

So,

$$\begin{aligned}\text{Bearing BA} &= 240^\circ 15' 30'' - 180^\circ 00' 00'' \\ &= 60^\circ 15' 30''\end{aligned}$$



In WCB the difference between FB and BB should be exactly 180°

$$BB = FB \pm 180^\circ$$

Use the +ve sign when $FB < 180^\circ$

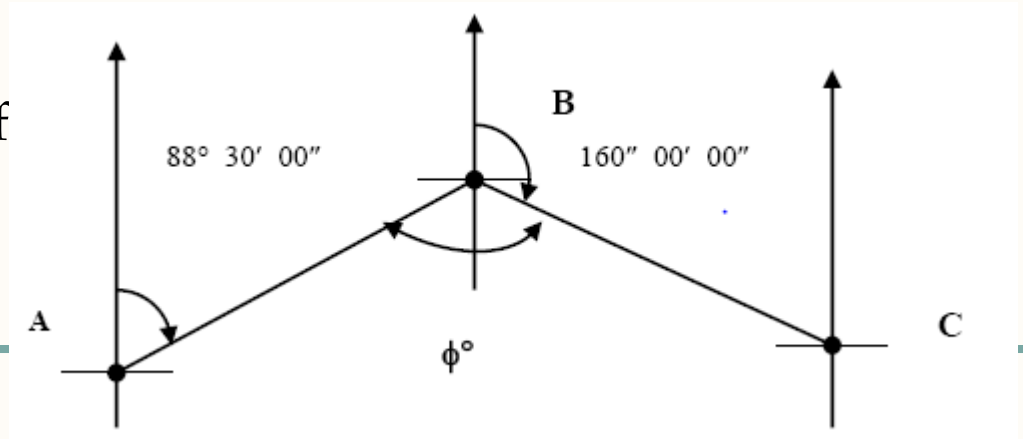
Use the -ve sign when $FB > 180^\circ$

EXAMPLE

Calculate the inner angle of B, if

Bearing AB = $88^{\circ} 30' 00''$

Bearing BC = $160^{\circ} 00' 45''$



Solution:

Bearing AB = $88^{\circ} 30' 00''$

So,

$$\begin{aligned}\text{Bearing BA} &= 88^{\circ} 30' 00'' + 180^{\circ} 00' 00'' \\ &= 268^{\circ} 30' 00''\end{aligned}$$

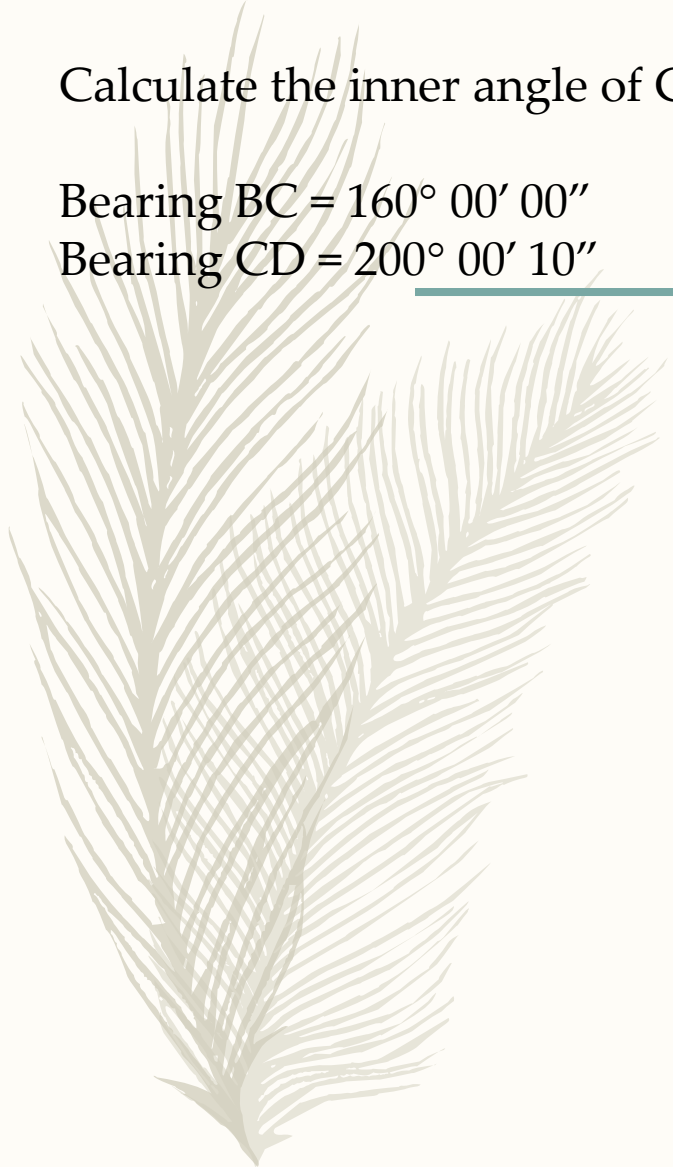
$$\begin{aligned}\angle B &= \text{Bearing BA} - \text{Bearing BC} \\ &= 268^{\circ} 30' 00'' - 160^{\circ} 00' 45'' \\ &= \underline{\underline{108^{\circ} 29' 15''}}\end{aligned}$$

EXERCISE

Calculate the inner angle of C, if :

Bearing BC = $160^{\circ} 00' 00''$

Bearing CD = $200^{\circ} 00' 10''$



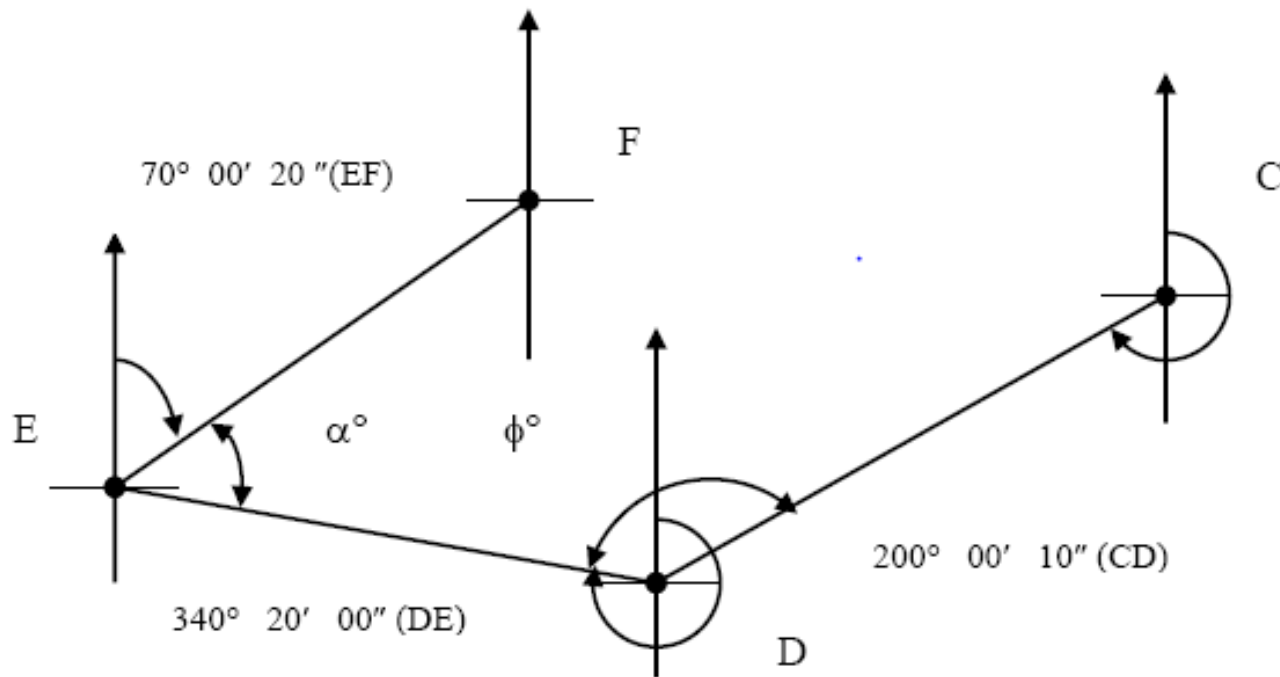
EXERCISE

Calculate the inner angle of D and E, if :

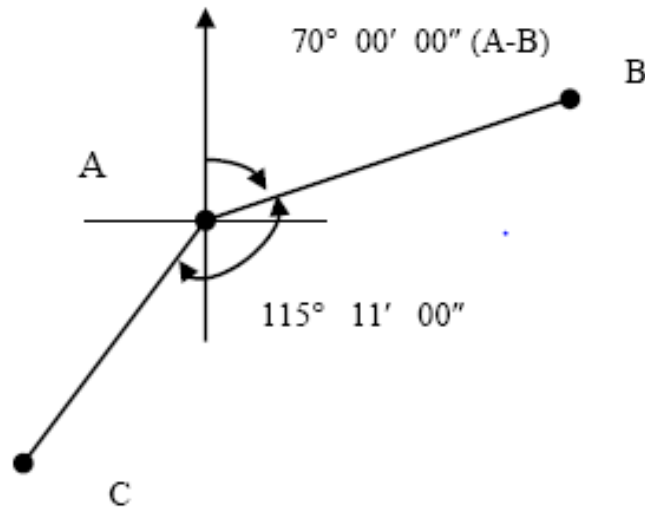
Bearing CD = $200^{\circ} 00' 10''$

Bearing DE = $340^{\circ} 20' 00''$

Bearing EF = $70^{\circ} 00' 20''$



BEARING FROM INTERNAL ANGLE AND A KNOWN BEARING.



EXAMPLE 1

Calculate line of BC, if bearing AB is $145^{\circ} 17' 20''$ and angle ABC is $86^{\circ} 45' 30''$

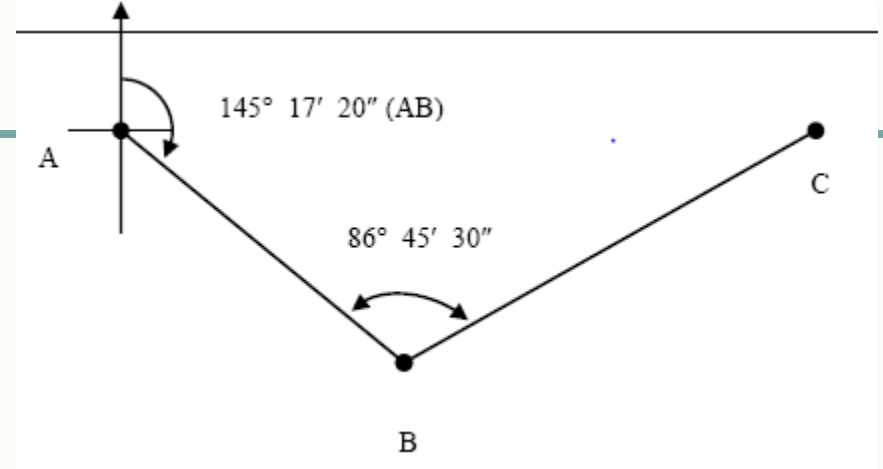
Solution:

$$\text{Bearing BC} = \text{Bearing BA} + \angle B$$

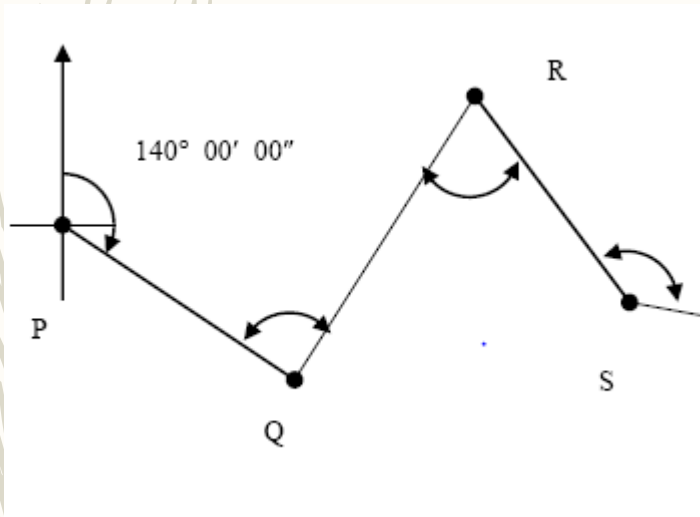
$$\begin{aligned}\text{Bearing BA} &= \text{Bearing AB} + 180^{\circ} \\ &= 145^{\circ} 17' 20'' + 180^{\circ} \\ &= 325^{\circ} 17' 20''\end{aligned}$$

$$\begin{aligned}\text{Bearing BC} &= \text{Bearing BA} + \angle B \\ &= 325^{\circ} 17' 20'' + 86^{\circ} 45' 30'' \\ &= 412^{\circ} 02' 50''\end{aligned}$$

$$\begin{aligned}\text{Bearing BC} &= 412^{\circ} 02' 50'' - 360^{\circ} 00' 00'' \\ &= \underline{\underline{52^{\circ} 02' 50''}}\end{aligned}$$



EXAMPLE 2



Bearing PQ = $140^{\circ} 00' 00''$

Sudut PQR = $092^{\circ} 15' 00''$

Sudut QRS = $121^{\circ} 13' 00''$

Calculate bearing of QR and RS

EXERCISE

Bering AB = $176^{\circ} 18' 30''$

Bering BC = $085^{\circ} 13' 40''$

Bering CD = $153^{\circ} 14' 30''$

Hitungkan sudut (θ°) dan sudut (ϕ°)

